

Mofiyinfoluwa Peter Orekoya

917-213-4364 | Fiyinorekoya001@gmail.com | [linkedin.com/in/forekoyacs](https://www.linkedin.com/in/forekoyacs) | [GitHub](https://github.com) | U.S. Citizen

EDUCATION

Purdue University

Bachelor of Science in Computer Science

- Minor in Microelectronics and Semiconductors

West Lafayette, IN

May 2028 (Expected)

EXPERIENCE

Georgia Tech Research Institute (GTRI) Applied Systems Lab

Software Engineering Intern

May 2026 – Aug 2026

Huntsville, AL

- Integrated and validated libraries and tooling across two U.S. Army air-and-missile-defense research platforms, resolving interface compatibility issues involving the **ABAL IDL**
- Built a GitLab CI test harness to benchmark throughput, CPU utilization, and memory consumption across deployment configurations and validate library and IDL changes
- Standardized build, test, lint, and formatting workflows with Pixi and CI across 5+ repositories, providing identical local and automated development checks

Purdue ROV (Remotely Operated Vehicle)

Software Team Member (OpenCV subteam)

Sept 2025 – Present

West Lafayette, IN

- Designed and implemented computer vision modules (Python, OpenCV, YOLOv8) for real-time crab detection, achieving **30+ FPS** inference with a **0.88 mAP@0.5** score
- Integrated CV outputs with the vehicle's embedded Linux software stack, supporting real-time video streaming
- Developed and tested camera integration layer across onboard hardware, handling device integration and compute constraints in an underwater robotics environment

ML@Purdue

Project Lead – Heterogeneous LLM Inference

Feb 2025 – Present

West Lafayette, IN

- Leading a Fall 2026 project investigating **CPU/GPU/NPU heterogeneous LLM inference**, focusing on speculative decoding with NPU drafting and GPU verification
- Directing architecture and experiments measuring latency, communication, synchronization, and accelerator-utilization tradeoffs

Intermediate Member – Kalshi Trading Team

Feb 2025 – May 2026

- Built and improved a TSA passenger-volume forecasting pipeline (**Python, XGBoost, Google Trends API**) for identifying mispriced Kalshi contracts, achieving **6.76 MAPE** with walk-forward validation

Purdue University Residences.

Resident Assistant (RA)

Aug 2026 – Present

West Lafayette, IN

- Supporting a residential community of 44 students by organizing weekly programs and connecting residents with university resources, opportunities, and support services

PROJECTS

Open-Source Distributed Computer Vision Runtime | Python, ZeroMQ, Protobuf

Jul 2026 - Present

- * Built a reusable **YAML-driven** distributed runtime for deploying camera and computer-vision workloads across heterogeneous Linux hosts, separating device capture, inference, configuration, and process supervision
- * Architected independently managed camera and CV processes communicating through **ZeroMQ** and **Protobuf**, with explicit registration, lifecycle, restart, and health semantics
- * Designed reliable camera-device discovery using persistent Linux device IDs and fallback paths, with pre-runtime validation of V4L2 capture modes and model/deployment compatibility

Custom 8-Bit CPU Simulator & Instruction Set Architecture | C, System Programming

Mar – Apr 2026

- * Designed a custom **8-bit RISC-style ISA**, including ALU operations, branching, load/store, and immediate-value encoding
- * Implemented a **functional CPU simulator in C** with register files, PC/memory model, and opcode and operand decoding to parse and execute machine code instructions
- * Built a **mini-assembler** to translate symbolic assembly into machine code

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, Java, SQL, R

ML/CV: PyTorch, OpenCV, YOLOv8, scikit-learn, NumPy, Pandas, RealSense SDK

Systems/Infrastructure : Linux/Unix, Docker, Git, ZeroMQ, Protobuf, Redis, Pixi, Artifactory, AWS, Podman